Project 1: Statistical Analysis of Machine Failure Risk

Assignment Type: Individual
Weighting: This project encompasses the learning and application aspects of both the Homework and Lab components (collectively 20%).
Learning Outcomes Assessed: LO1, LO3
Topic Coverage: Weeks 1-7 (Descriptive Statistics, Probability Distributions, Sampling Distributions, Hypothesis Testing, Confidence Intervals)

Objective

To develop statistical computing skills for performing a complete statistical analysis on a given engineering dataset. This includes data visualization, descriptive statistics, probability distribution fitting, sampling distributions, and inferential statistics (confidence intervals and hypothesis testing) using Excel or MATLAB.

Problem Statement

You are an engineering intern in a manufacturing company. The maintenance department has provided you with a dataset (`machine_failure`) collected from industrial machines, including sensor readings and failure risk indicators. Your task is to analyze this data to gain meaningful insights into the system's operating behavior and to assess whether the observed performance is consistent with acceptable operating conditions.

Dataset

The dataset `machine_failure` will be provided on Moodle. It contains 1,000 observations from industrial machines and six columns: `Temperature` (the operating temperature), `Pressure` (the operating pressure), `Vibration` (the vibration level), `Failure_Risk` (the failure risk indicator: 0 = No risk, 1 = Risk), `Machine_Type` (the type of machine), and `Shift` (the operating shift: Day or Night).

Tasks

1. **Data Import and Visualization (LO3)**
   - Import the dataset into your chosen software (Excel or MATLAB).
   - Select **one continuous variable** (e.g., Temperature or Vibration) and create the following plots to visualize its distribution:
     - A histogram with an appropriate number of bins.
     - A boxplot to identify any potential outliers.

        - A normal probability plot (QQ plot) to visually assess if the data follows a normal distribution.

2. **Descriptive Statistics (LO1, LO3)**
    - Calculate and display the following descriptive statistics for selected variable:
        - Mean, Median, and Mode.
        - Standard Deviation, Variance, and Range.
        - First and Third Quartiles, and the Interquartile Range (IQR).
    - Based on your boxplot, list any data points that could be considered outliers (using the 1.5 * IQR rule).
    - **Interpretation (Written Answers)**
        - What does the boxplot show about skewness and outliers?
        - Describe the distribution using the plots.
3. **Probability and Fitting Distributions (LO1, LO3)**

    PART A: Probability & Random Variables

    - Define: Sample space of selecting 2 machine records.
        - Compute probabilities: You will estimate probabilities by looking at how often certain conditions appear in the data. For each event, count how many rows in the dataset satisfy the condition and divide by the total number of rows. This gives an estimate of the probability based on real observations.
    - Events: A = “Temperature >50”, B = “Failure Risk = 1”
      Compute:
        a) $P(A \cap B)$
        b) $P(A \cup B)$
        c) $P(A \setminus B)$

    PART B: Probability Distribution Fitting

    - Assume the selected variable follows a normal distribution.
    - Estimate the parameters (mean and standard deviation) of this normal distribution from your data.
    - Overlay the fitted normal distribution's probability density function (PDF) on your histogram from Task 1 to visually compare the fit.
    - **Explain:** Why is the Normal distribution suitable (or not suitable) for this variable?
4. **Sampling Distributions (LO3)**
    - Take random samples of sizes n = 5, 20, and 50 from the Temperature variable.
    - For each n, repeat the sampling many times (e.g., at least 100) and compute the sample mean.
    - Plot the sampling distribution of the sample mean for each n.

- Interpretation (Written Answer)
    - Describe how the shape and spread of the sampling distribution change as n increases.

5. **Confidence Interval Estimation (LO1, LO3)**
    - Calculate a 95% confidence interval for the population mean of the selected variable (e.g., Temperature or Vibration).
    - Calculate a 95% confidence interval for the population variance of the selected variable.
6. **Hypothesis Testing (LO1, LO3)**
    - The system specification requires that the mean power usage be at most 10 units. Conduct a one-sample hypothesis test at a significance level of **α = 0.05 to** determine whether there is sufficient evidence to conclude that the population mean meets this requirement
        - Clearly state your null and alternative hypotheses.
        - Perform the appropriate t-test using your chosen software and display the p-value.
        - State your conclusion in the context of the problem (i.e., does the system meet the power usage requirement?).

Deliverables

- A single, well-organized Excel file or MATLAB live script that performs all the tasks. (`Project1_YourStudentID.xlsx` or `Project1_YourStudentID.mlx`)
- A brief summary of your main conclusions, written at the end of the file (as comments or notes, depending on the software used).

Generative AI Encouragement (as per Section 5.2)

You are permitted to use GenAI tools like ChatGPT to help you with this assignment.

- **How to use:** You may use it to generate code snippets for specific tasks (e.g., "Write MATLAB code to create a normal probability plot" or “Write Excel formulas to compute a confidence interval”), to debug error messages, or to explain statistical concepts (e.g., "Explain the difference between a t-test and a z-test").
- **Acknowledgment:** At the beginning of your file (Excel or MATLAB), include a brief statement acknowledging which GenAI tool you used and for what purpose (e.g., "% GenAI tool (ChatGPT-3.5) was used to help generate the code for the histogram and to debug the QQ plot function.").
- **Critical Thinking:** Remember that you are responsible for the final analysis and conclusions. Verify the GenAI output against your lecture notes and ensure you understand every line of code in your final submission.